Back

Knowledge Check: Hidden Markov Models (HMMs)
Practice Assignment • 15 min

Due Oct 21, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is an example of a **domain-specific feature** in CRF-based NER? 1 / 1 point

- Whether the word is at the beginning of the sentence
- Whether the word appears in a known list of state names
- Token length in characters
- Whether the word contains the letter 'e'

Nice work

Correct! Domain-specific lists can greatly help CRF models.

2. What type of model is a CRF? 1 / 1 point

- A visual model
- A translation model
- A discriminative model
- A generative model

Nice work

Correct! CRFs are discriminative models.

3. Which of the following are components of a Markov chain model? 1 / 1 point

- [ ] N-gram token frequency table
- [x] Set of states

Nice work

Correct! A Markov chain consists of a set of states.

- [x] Transition probabilities between states

Nice work

Correct! Transition probabilities are a key component of the model.

- [ ] Parts of speech tags

4. Which type of feature can be used in a CRF? 1 / 1 point

- Only word length
- Only punctuation symbols
- Only word frequency
- Orthographic, morphological, linguistic, and domain-specific features

Nice work

Correct! CRFs use a rich set of feature types.

5. What are the three components of an HMM? 1 / 1 point

- Tokens, states, word embeddings
- Hyperparameters, learning rate, regularization
- Hidden states, transition probabilities, and emission probabilities
- Hidden states, grammar rules, sentence templates

Nice work

Correct! These are the core components of an HMM.

6. What is required to compute the probability of a sequence in a Markov chain? 1 / 1 point

- The number of states visited
- The initial state probability and the transition probabilities
- The length of each word in the sequence
- The number of tokens with punctuation

Nice work

Correct! The sequence probability depends on the initial state probability and transition probabilities.

7. What is the key difference between Markov chains and hidden Markov models (HMMs)? 1 / 1 point

- Markov chains cannot be used in NER
- HMMs only process visual data
- Markov chains can model future dependencies
- HMMs have hidden states that emit observed tokens

Nice work

Correct! HMMs involve hidden states generating observable output.

8. What is a key **drawback** of using HMMs for NER? 1 / 1 point

- They cannot model the BIO label format
- They require handcrafted rules for every label
- They make strong independence assumptions that limit modeling of long-range dependencies
- They cannot process multiple sentences

Nice work

Correct! The strong independence assumption is a drawback of HMMs.

9. What is one major advantage of using CRFs for NER? 1 / 1 point

- They operate faster than all neural networks
- They use fewer training examples
- They model long-range dependencies and arbitrary features
- They eliminate the need for features

Nice work

Correct! CRFs model long-range dependencies and arbitrary features.

10. Which of the following are advantages of CRFs over HMMs? 1 / 1 point

- [ ] Fully unsupervised training
- [x] Better handling of long-range dependencies

Nice work

Correct! CRFs outperform HMMs in modeling long-range dependencies.

- [x] Ability to use arbitrary features, including past and future observations

Nice work

Correct! CRFs can use arbitrary features.

- [ ] Faster tokenization of input text

11. What is the Markov property? 1 / 1 point

- The ability to use all past states to predict the next word
- The ability to process images with text
- The future state depends only on the current state, not on past states
- The ability to generate summaries from text

Nice work

Correct! The Markov property states that the next state depends only on the current state, not on the history.

12. In a Markov chain, what does a "state" represent? 1 / 1 point

- A sentence boundary
- A position in a sequence with possible transitions to other positions
- A random number generator
- A fixed punctuation mark

Nice work

Correct! States represent positions in a sequence with transitions to other states.